

교육용 자료 안내

본 자료는 Darun Color Institute 의 색채 교육 과정을 위해 제공되는 교육용 문서입니다. 학생들이 안전한 재료 취급을 학습하기 위해 사용된 원료의 물리화학적 특성과 안전 정보를 공개합니다.

다만, 이 자료에 포함된 정보는 교육 목적으로만 활용되어야 하며, 상업적 목적의 제품 제조 및 판매는 엄격히 금지됩니다.

화장품 제조업 허가를 취득한 자가 독립적으로 제품을 개발·판매할 경우, 다음을 반드시 준수해야 합니다:

- 신뢰할 수 있는 화장품 원료 전문 제조사로부터 원료를 직접 확보
- 해당 제조사로부터 최신 MSDS 및 시험성적서(CoA, 안정성 시험 등)를 개별 요청
- 식품의약품안전처 규정에 따른 원료 검증 및 제품 신고

이는 소비자 안전과 법적 책임을 보장하기 위한 필수 절차입니다.

Educational Material Notice

This document is provided by Darun Color Institute as an educational resource for our color science training programs. To support student learning in safe material handling practices, we disclose the physical, chemical, and safety characteristics of the raw materials used in our formulations.

However, all information contained in this material is strictly for educational purposes only. Any commercial use for product manufacturing or sales is strictly prohibited.

Should you obtain cosmetic manufacturing authorization and independently develop products for commercial purposes, you must:

- Source raw materials directly from reliable cosmetic ingredient manufacturers
- Request current MSDS and certificates of analysis (CoA) individually from the respective manufacturers
- Verify all raw materials and complete product registration in compliance with Korea Food and Drug Administration (KFDA) regulations

These procedures are mandatory requirements to ensure consumer safety and fulfill legal responsibilities.

Material Safety Data Sheet (MSDS)

1. Chemical Product and Company Identification

Item	Information
Product Name	Educational Color TiO2-WHITE
Recommended Use	Cosmetic ingredient
Means of Identification	Not available
Relevant identified uses	Not applicable

2. Hazards Identification

Category	Details
GHS Classification	Health: H351, H372
Hazard Statements	H351: Suspected of causing cancer H372: Causes damage to organs through prolonged or repeated exposure
Precautionary Statements	P201: Obtain special instructions before use P202: Do not handle until all safety precautions have been read and understood P260: Do not breathe dust/fumes/gas/mist/vapours/spray P264: Wash thoroughly after handling P270: Do not eat, drink or smoke when using this product P280: Use personal protective equipment as required P308+P313: If exposed: Call a POISON CENTER or doctor/physician P314: Get medical advice/attention if you feel unwell P405: Store locked up P501: Dispose of contents/container according to regulations

3. Composition/Information on Ingredients

**Composition ratios are provided as approximate ranges for educational purposes*

Substance/mixture : Mixture

There are no ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment, are PBTs or vPvBs or have been assigned a workplace exposure limit and hence require reporting in this section.

4. First Aid Measures

Route of Exposure	First Aid Procedure
Eye Contact	Rinse with water for at least 20 minutes. Seek medical attention immediately.

Skin Contact	Rinse with water for at least 20 minutes. Remove contaminated clothing. Seek medical attention if irritation persists.
Inhalation	Move to fresh air immediately. If breathing is difficult, provide oxygen. Seek medical attention.
Ingestion	Do not give anything by mouth to an unconscious person. Seek medical attention immediately.

5. Fire-Fighting Measures

Item	Information
Suitable Extinguishing Media	Small fire: Dry sand, dry chemical, alcohol-resistant foam, water spray, general foam, CO2 Large fire: Water spray/mist, general foam
Specific Hazards	Container may explode when heated. Fire may generate irritating or toxic gases.
Special Protective Equipment	Move container from fire area if not dangerous. Cool containers with water even after extinguishing.

6. Accidental Release Measures

Action	Description
Prevention	Eliminate all ignition sources. Prevent dust formation. Ventilate contaminated area.
Handling Spills	Small spills: Absorb with sand or non-combustible material and place in container Large spills: Make a ditch away from liquid spills. Use clean shovel to place spillage into clean, dry container.
Environmental Protection	Prevent entry into waterways, sewers, basements, and confined spaces. Cover powder spills with plastic sheet to prevent diffusion.

7. Handling and Storage

Category	Requirements
Handling	Work in well-ventilated area Do not breathe dust or spray Wash thoroughly after handling Do not eat, drink or smoke when using product Avoid high temperatures
Storage	Store in cool dry place below 20°C Keep container tightly closed Store locked up Prevent dust formation Store away from ignition sources and incompatible materials

8. Exposure Controls/Personal Protection

Control Type	Requirement
Control Parameters (CI 77891)	TWA: 10 mg/m ³ (Carcinogenicity Level 2)
Engineering Controls	Process isolation, use local exhaust ventilation, or keep air level below exposure limit
Respiratory Protection	Wear respiratory protection certified for particulate matter (face filter type dust mask or air filter type with high efficiency particulate filter)
Eye Protection	Wear safety glasses or goggles
Hand Protection	Wear appropriate protective gloves considering chemical properties
Body Protection	Wear appropriate protective clothing considering chemical properties

9. Physical and Chemical Properties

Property	Value
Appearance	White paste
Odor	Odorless
pH	No data
Flash Point	Not applicable
Flammability	Can be ignited by heat, sparks or flames
Solubility	No data

10. Stability and Reactivity

Item	Information
Chemical Stability	Stable under normal temperature and pressure conditions
Conditions to Avoid	Heat, sparks, flames, and ignition sources
Incompatible Materials	Combustible materials
Hazardous Decomposition	Fire may generate irritating or toxic gases. Some vapors may cause dizziness and suffocation.

11. Toxicological Information

Effect	Description
Effect	Description
Acute Toxicity (Oral - CI 77891)	LD50 > 2000 mg/kg (Mouse)

Acute Toxicity (Inhalation - CI 77891)	LC50 > 100 mg/L 96hr (No death observed)
Skin Irritation (CI 77891)	Skin irritation test (Rabbit): No irritation
Eye Irritation (CI 77891)	No irritation
Respiratory Irritation	Inhalation of substance may be harmful. Some vapors may cause dizziness and suffocation.
Carcinogenicity (CI 77891)	Level 2 (Suspected of causing cancer - Powder form containing 1% or more of TiO ₂ particles ≤ 10µm)
Long-term Exposure	Causes damage to organs through prolonged or repeated exposure

12. Ecological Information

Category	Information
Fish Toxicity (CI 77891)	LC50 > 6.82 mg/L (Rat, OECD TG 403)
Crustacean Toxicity (CI 77891)	LC50 > 500 mg/L 48hr (Daphnia magna)
Algae Toxicity (CI 77891)	EC50 > 50 mg/L 72hr (Selenastrum capricornutum)
Persistency and Degradability	No data available
Bio-condensability	No data available

13. Disposal Considerations

Item	Information
Disposal Method	Dispose of contents and container in accordance with regulations specified in the Waste Management Act

14. Transport Information

Item	Information
Classification	Non-hazardous for air transport (IATA regulation)
UN Number	Not applicable
Proper Shipping Name	Not applicable

15. Regulatory Information

Item	Information
Regulatory Basis	Prepared in accordance with Article 41 of the Occupational Safety and Health Act and relevant KFSA regulations

16. Other Information

Item	Information
Disclaimer	To the best of our knowledge, the information contained herein is accurate. However, we cannot accept any responsibility or liability for any consequences arising from use of it. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution.
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